

NBT & DAB Staining

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Purpose. Detection of superoxide (NBT) or hydrogen peroxide (DAB).

Reagent / condition	Preparation or use
Bleaching solution	Ethanol : lactic acid : glycerol = 3:1:1
25 mM HEPES	pH 7.6; adjust with KOH; prepare using 1× PBS as noted
NBT solution	25 mM HEPES, pH 7.6 + 0.01% Triton X-100 + 0.01% NBT
DAB solution	D.W. + 0.01% Triton X-100 + 0.1% DAB, pH 5.8
50 mL NBT	50 mL HEPES + 25 µL 20% Triton + 5 mg NBT
50 mL DAB	50 mL HEPES + 25 µL 20% Triton + 50 mg DAB

Fresh reagent Prepare NBT and DAB staining solutions on the day of use. When adding Triton X-100 from a 20% stock, calculate the volume from the final stock concentration.

NBT workflow

1. Vacuum-infiltrate inflorescences with NBT solution.
2. Incubate for 3 h in the dark.
3. Transfer to bleaching solution until chlorophyll is completely removed; 70 °C for 1 h is recommended where overnight room-temperature bleaching causes diffusion.

DAB workflow

1. Vacuum-infiltrate inflorescences with DAB solution.
2. Incubate for 6 h in the dark.
3. Bleach as above until chlorophyll is removed.

Root NBT alternative Stain roots for 15 min in 2 mM NBT in 20 mM phosphate buffer, pH 6.1. Stop the reaction by transfer to distilled water.

Source: [Root NBT reference](#)